1/15/2019

Lecture 1

Q. Why do we need an advanced propulsion system?

→ to fly faster, higher & cheaper.

→ Current high-performance LBR turbofans are limited to $M \approx 1.5$; $h_{ceil} = 45$ kft

→ Extreme exemplar is SR-71, whose speed is $M = 3.2$ @ 80 kft

→ Ramjets allow higher flight Mach #s

Generally, these are used in rocket boosted missiles, capable of Mach 4+. Some limited use in 50s & 60s in manned propulsion

Now, ramjets are used almost exclusively in missiles. Still limited to $M \leq 6$.

Q. Why cheaper?

Current cost to place something in orbit (LEO) ~ 200 km altitude is about \$10k/lb.

→ Chemical rockets (liquid & solid) are the only means to achieve orbit today.

→ They are expensive b/c

- rockets operate w/ very small margins (they are not robust or flexible)
- They require a large standing army of ppl. to support them

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→ Typically are not reusable, and if they are reusable, refurbishment costs are expensive

→ 65% of mass on rocket is oxidizer.

(would rather get for free from atm.)

To reduce cost, we want to make vehicles more "aircraft-like"

- robust
- no expensive reusing costs or time delay for reuse.
- want to use O_2 from atm. for some fraction of the flights

Possible options for advanced propulsion system include scramjet & pressure gain system (PDE, RDE)

For example, to achieve LEO, need to accelerate a payload to $M \sim 25$

One scenario would be:

- use turbojet/fan to power airplane-like vehicle from takeoff to $M \approx 1.2$, 35 kft
- Engine would transition to ramjet (all compression achieve via ram) and accel. to $M \approx 5$, 75 kft
- @ $M = 5$, burn the fuel supersonically (scramjet mode) until $M \sim 10-12$ & ceil ≥ 100 kft
  Combustion Mach # $\approx 1.2 - 2.5$
- @ 100 kft, altitude is so high that density is too low to sustain combustion. So, convert to rocket mode.

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